

Lecture 01

Introduction to Linear Algebra and Data Science

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Part I Relation of Data and Linear Algebra

Why Data are Important



Data are the new “oil”!

数据是新的石油

Oil can drive cars, airplanes, cellphones, ...

Hardware

Data can drive AI and decision making,

“Software”



The Economist

Menu

Weekly edition

The world in brief

🔍 Search ▼

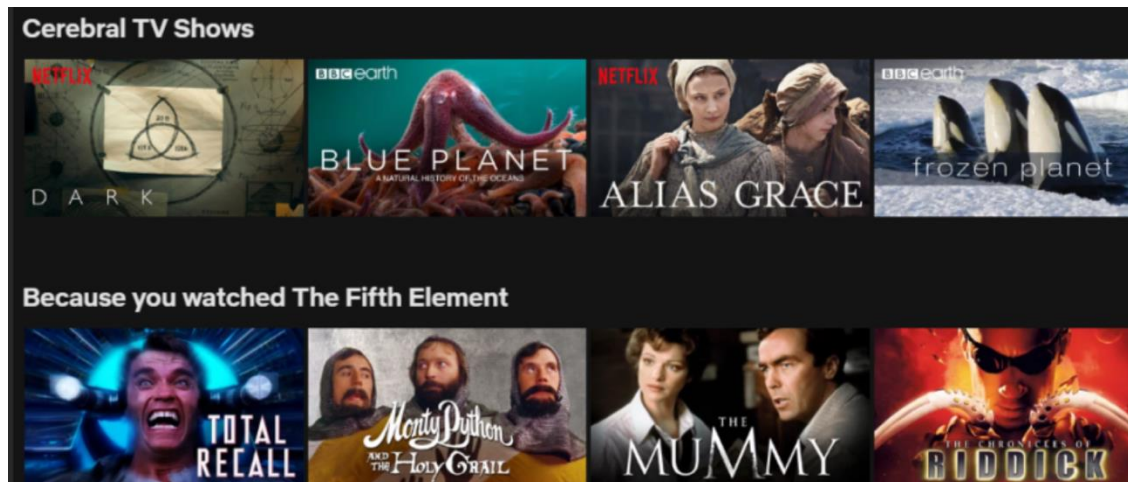
Leaders | Regulating the internet giants

The world's most valuable resource is no longer oil, but data

The data economy demands a new approach to antitrust rules

2017, Economist article

What are “Data”?



First Challenge: How to “Express” them?



Figure?

$$\begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix}$$

vector!

First Challenge: How to “Express” them?



Figure?

157	153	174	168	150	152	129	151	172	161	165	166
156	182	163	74	75	62	33	17	110	210	180	164
180	180	50	14	34	6	10	33	48	106	169	181
206	109	5	124	191	111	120	204	166	15	56	180
194	68	197	251	237	239	239	228	227	67	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	165	252	236	231	149	178	228	43	95	234

Matrix!

Part II Brief Intro of Linear Algebra

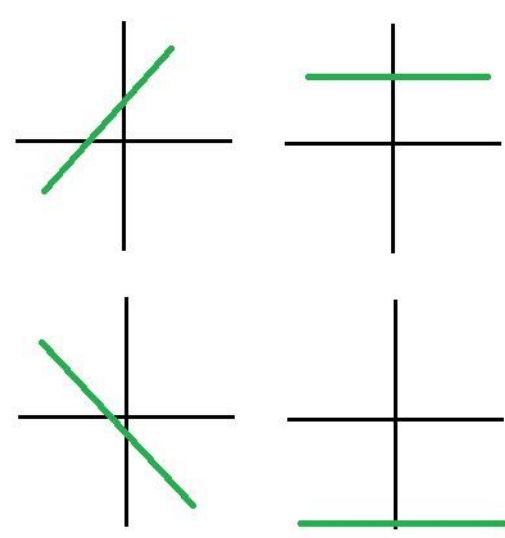
What is Linear Algebra?

Algebra:

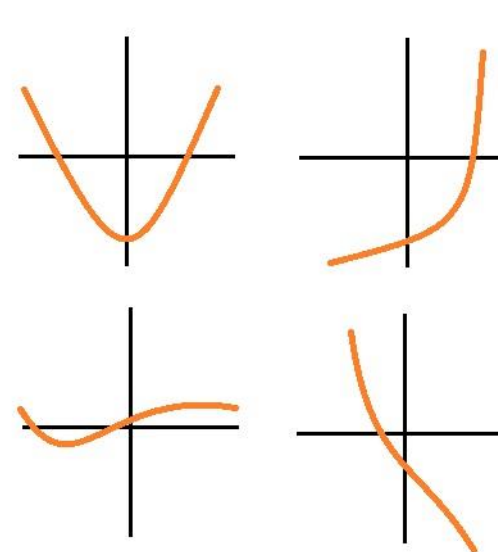
Mathematical representation of
subjects and **operations**

$X, Y, x, y, A, B, \dots$ addition +
subtraction -
multiplication $\times$

Linearity:



linear functions



non-linear functions

2-D representation

What is Linear Algebra?

Primary/High-School Algebra

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

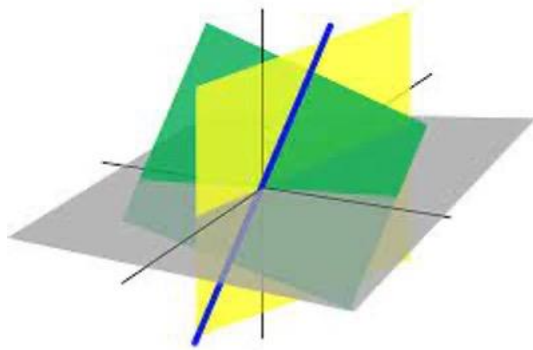
Linear Algebra

The diagram shows the equation $Ax + By = C$ with the following labels and arrows:

- Coefficient of x** (blue) with an arrow pointing to A
- Coefficient of y** (blue) with an arrow pointing to B
- Variables** (orange) with arrows pointing to x and y
- Constant** (black) with an arrow pointing to C

Standard form of an Linear Equation

What is **Linear Algebra**?



Mathematically, **vector spaces**
and linear transformations

Applications

- **Machine learning and data science**
- Computer vision and graphics
- Graph theory
- Control theory
- Cryptography
- Fractals and chaos
- Energy systems
- Network systems
- Genetics
- Etc. ...

Practically, it can be applied to **any** problems with
vector, matrix-type data, and linear models

Why Need Linear Algebra? A1: Fundamental

Quora

Search for questions, people, and topics

What exactly is linear algebra? Why do we need it?

Why study linear algebra?

Ask Q

Asked 9 years, 8 months ago Modified 3 years, 4 months ago Viewed 133k times



128

Simply as the title says. I've done some research, but still haven't arrived at an answer I am satisfied with. I know the answer varies in different fields, but in general, why would someone study linear algebra?

<https://math.stackexchange.com/questions/256682/why-study-linear-algebra>

Linear algebra is **beyond important**, it is **fundamental** to so many fields that I cannot count them all.

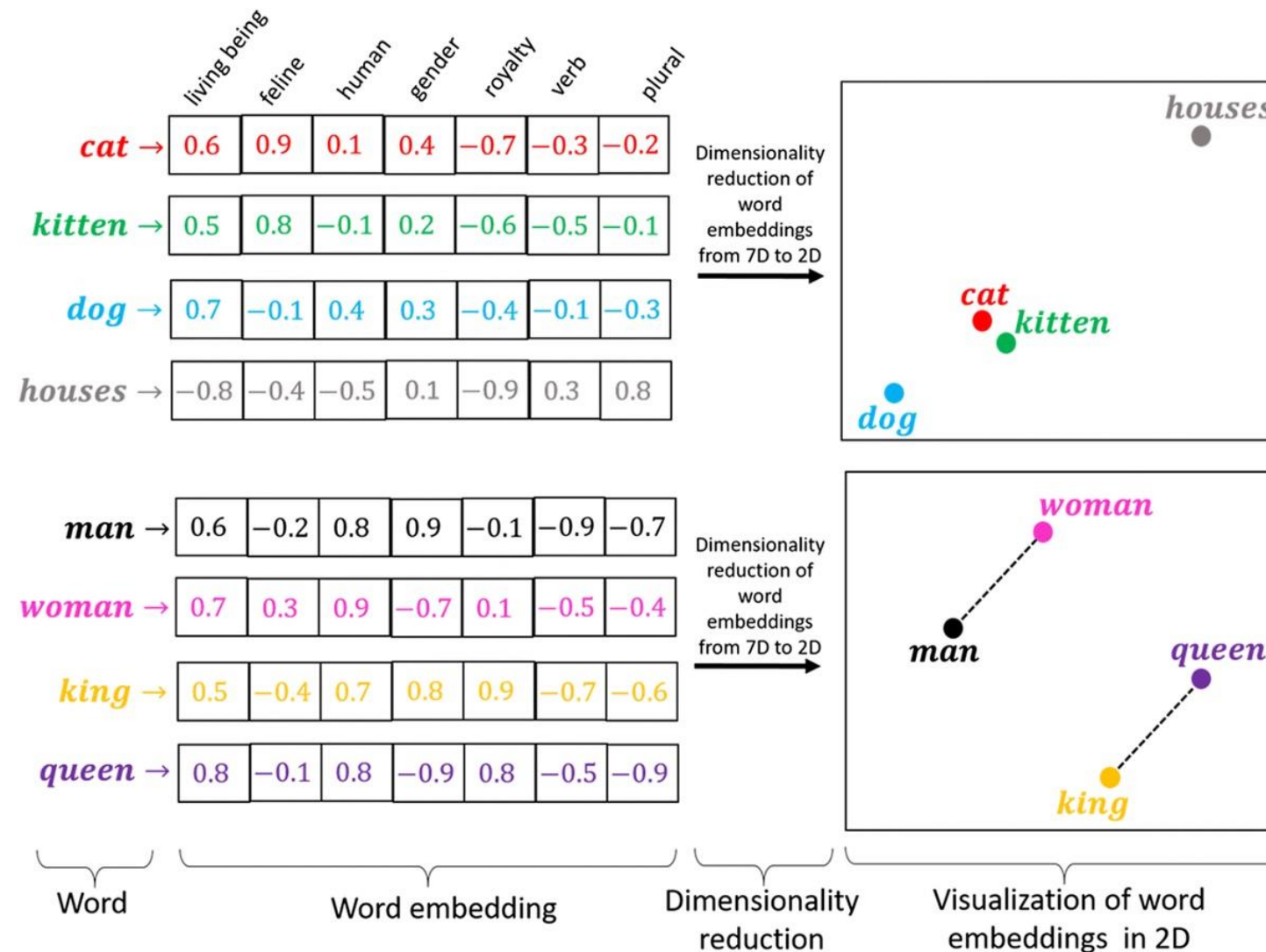
Having studied Engineering, I can tell you that Linear Algebra is fundamental and an extremely powerful tool in **every single** discipline of Engineering.

3D computer graphics? Linear algebra.
Quantum mechanics? Linear algebra.
Weather forecast models? Linear algebra.

Study it if you are into economics, computer graphics, physics, chemistry, statistics or anything quantitative (in today's world, that's everything).

Example in AI

Word Embedding: transform words to vectors (Mikolov of Google, 2004)

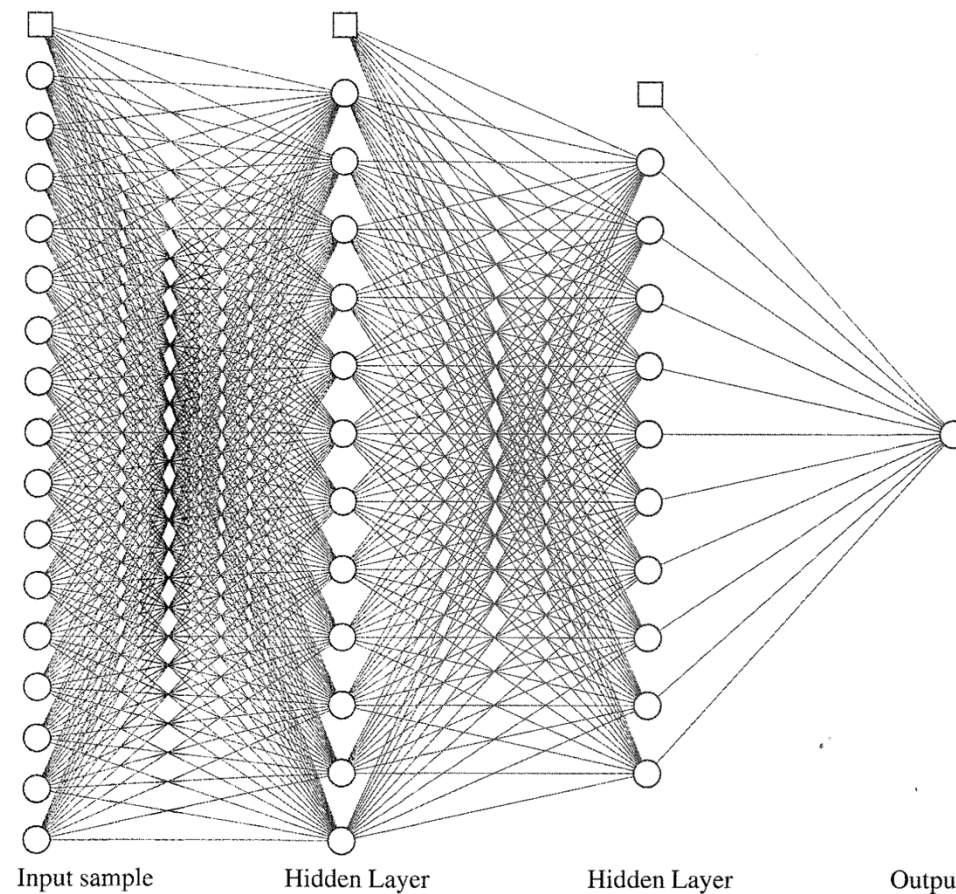


(Figure resource:
<https://medium.com/@hari4om/word-embedding-d816f643140>)

When you talk to chatGPT/deepseek,
words are transformed to **vectors** !

Example in AI: Neural Network

Neural Network



(Figure resource: linear algebra and learning from data by Gilbert Strang)

- A neural network is (almost) a collection of **vectors** and **matrices**
- **Example of neural nets:** GPT4, deepseek, Gemini 3,...

Nvidia's Success: Betting on Linear Algebra Computation

Why is everyone suddenly talking about **Nvidia**, the nearly **\$4 trillion-dollar** company fueling the AI revolution?

Intel and AMD: CPU guru
Dominate Chip industry for a few decades

Nvidia, graphics processing units (GPUs)
--designed for rendering images
i.e. for matrix computation, much better than CPUs
(say, 100X times faster)
--so GPUs are great for neural network computation

This Course

It is a math course.

Mathematical definitions, lemmas, theorems, and formal expressions will be marked in a colored box..

But not a “pure math”-type course

Skipped some abstract math results compared to a math course

More data science problems will be demonstrated to motivate the concepts (compared to classical Linear Algebra courses)

For instance, regression problems, graph matrices, searching, etc. (will cover as much as possible)

--This is strongly advocated by Steven Boyd's book and course in Stanford

Part III Advice on Learning Linear Algebra

Strategies for Effective Math Learning: General

Practice Regularly

Math is learned by doing.

--Solve as many practice problems in the books you can find

Review Consistently

--formal review: biweekly or weekly

--informal review:

just recall what you learned, during lunch or walking

Collabarative Learning

Discuss with Peers

Encourage forming study groups to share understanding and problem-solving strategies.

(don't copy homeworks from others)

Teaching Peers

"Teaching others helps reinforce your own understanding."

Ask and answer questions on Piazza

(or other platfroms we decided to use)

Active Learning

Draw mind-map & write summary yourself.

--Ask yourself: what did I really learn?

Link Math to Real-Life Applications

--I gave examples of real-life applications,
you can try to identify more

Refer to Multiple books

--Every author explains concepts differently
—explore what resonates with you.

General Goal

Classic linear algebra training:

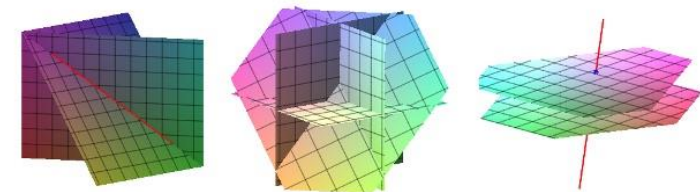
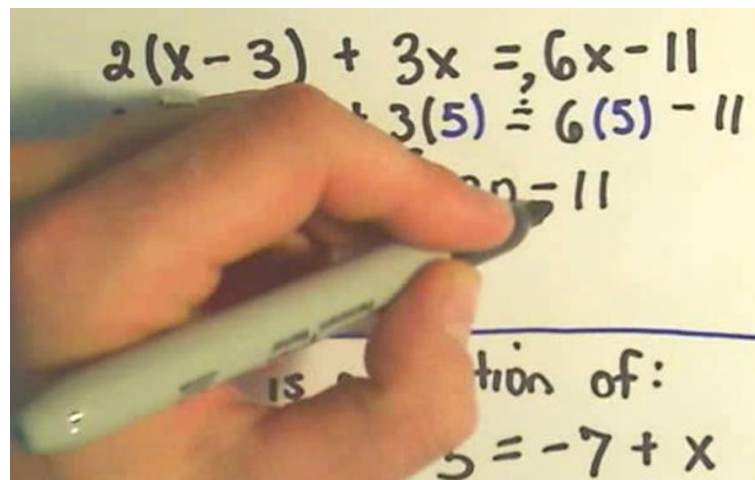
How to solve linear equations as fast as possible as humans?

Computers will do the job!

(Unfortunately, in your exams, you sometime need to solve linear equations by hands; just occasionally!)

Understand Concepts, Not Just Procedures

Learn **How** and **what**, and also **Why** (keep asking yourself why



Part IV Intro to Vectors

Basic Components in Linear Algebra

Question1: Basic components in Linear Algebra?

Vectors, matrices, and their operations etc

Question2: What are the most fundamental component in Linear Algebra?

Vectors!

Vectors

Definition (column vectors)

$$\mathbf{v} = [v_1, v_2, \dots, v_i, \dots, v_n]^T \xrightarrow{\text{Transpose}}$$

$\downarrow$ column vector $\downarrow$ entry/element

$$= \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \equiv \underbrace{(v_1, \dots, v_n)}$$

also written as

Convention: vectors considered as columns

Examples of Vectors

Example:

Zero Vector $\mathbf{v} = (0,0,0,\dots,0)$

One Vector $\mathbf{v} = (1,1,1,\dots,1)$

Warning:

--In math operation,
A number is a vector;
--In coding, a number
cannot be treated as a vector often.

$v = (2)$ is also a vector with size 1.

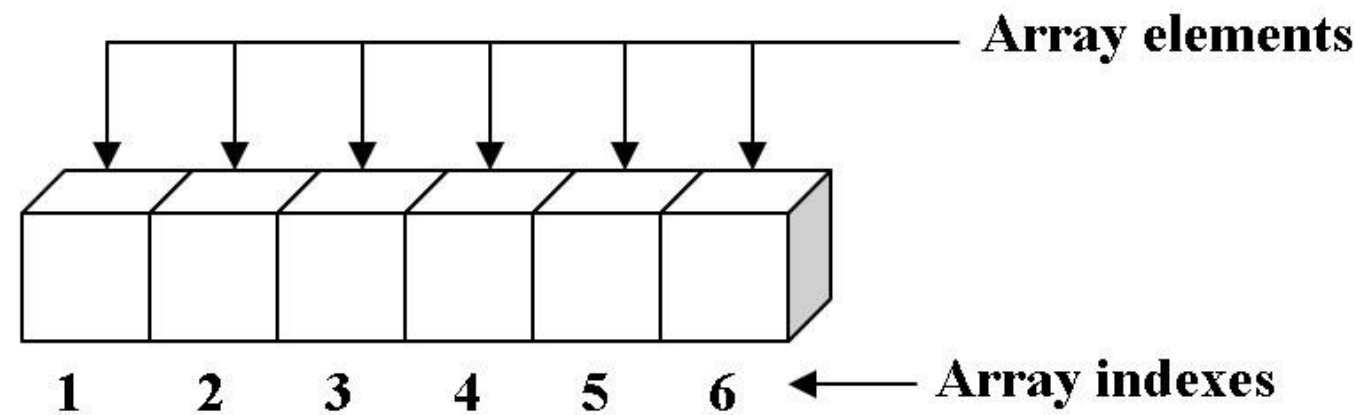
So you can say “a number is also a vector”.

We use the term **scalar** to denote a single number, distinguishing it from a **vector**, which is an array of multiple numbers.

Examples of Vectors

Example:

An **array** data structure in computer algorithms



One-dimensional array with six elements

Examples of Vectors

Example:

Color **R****G****B** Vectors



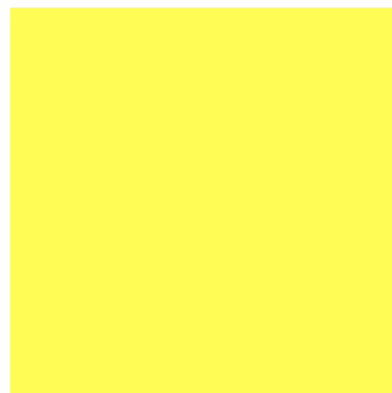
$(1, 0, 0)$



$(0, 1, 0)$



$(0, 0, 1)$



$(1, 1, 0)$



$(1, 0.5, 0.5)$

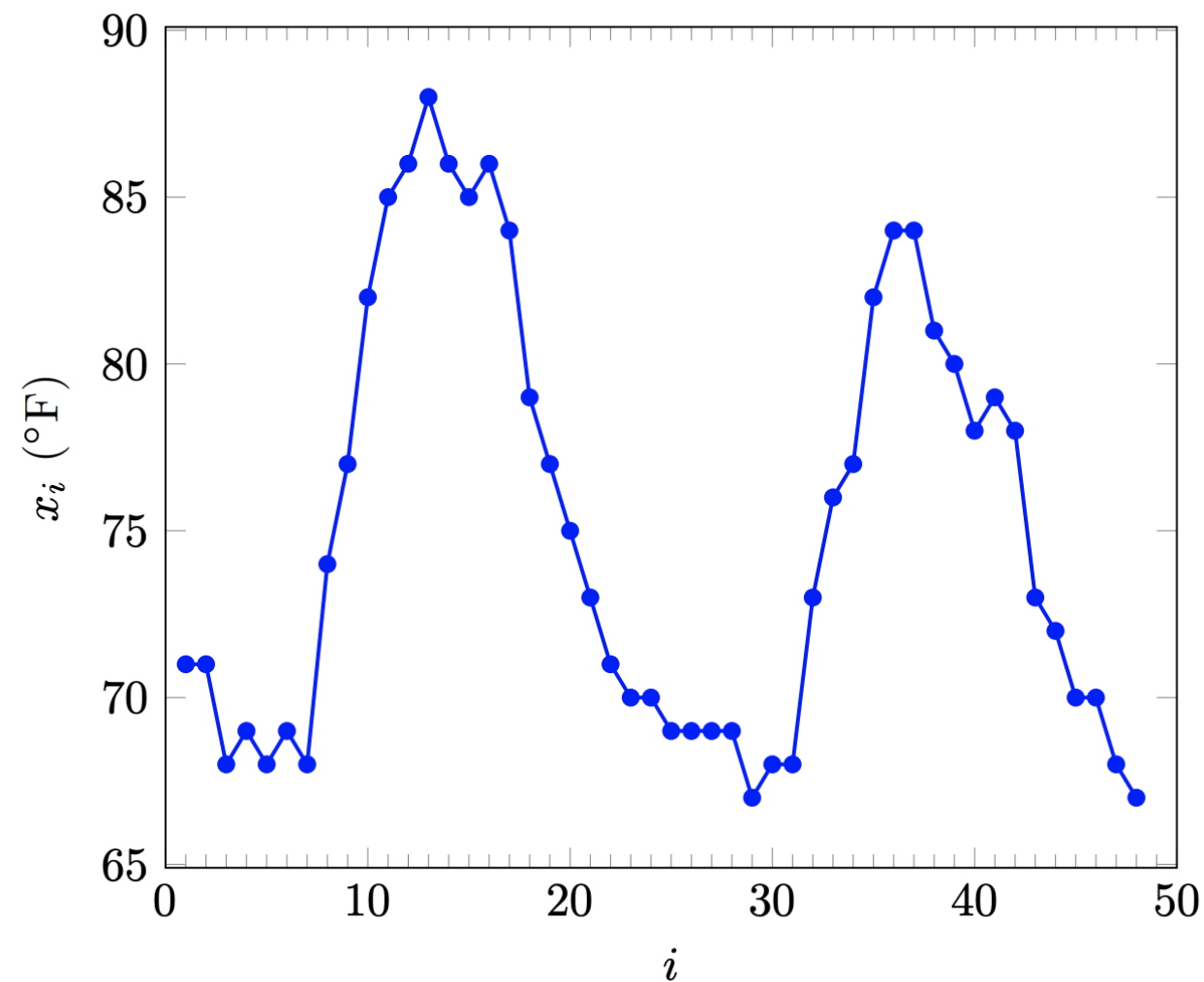


$(0.5, 0.5, 0.5)$

Examples of Vectors

Example:

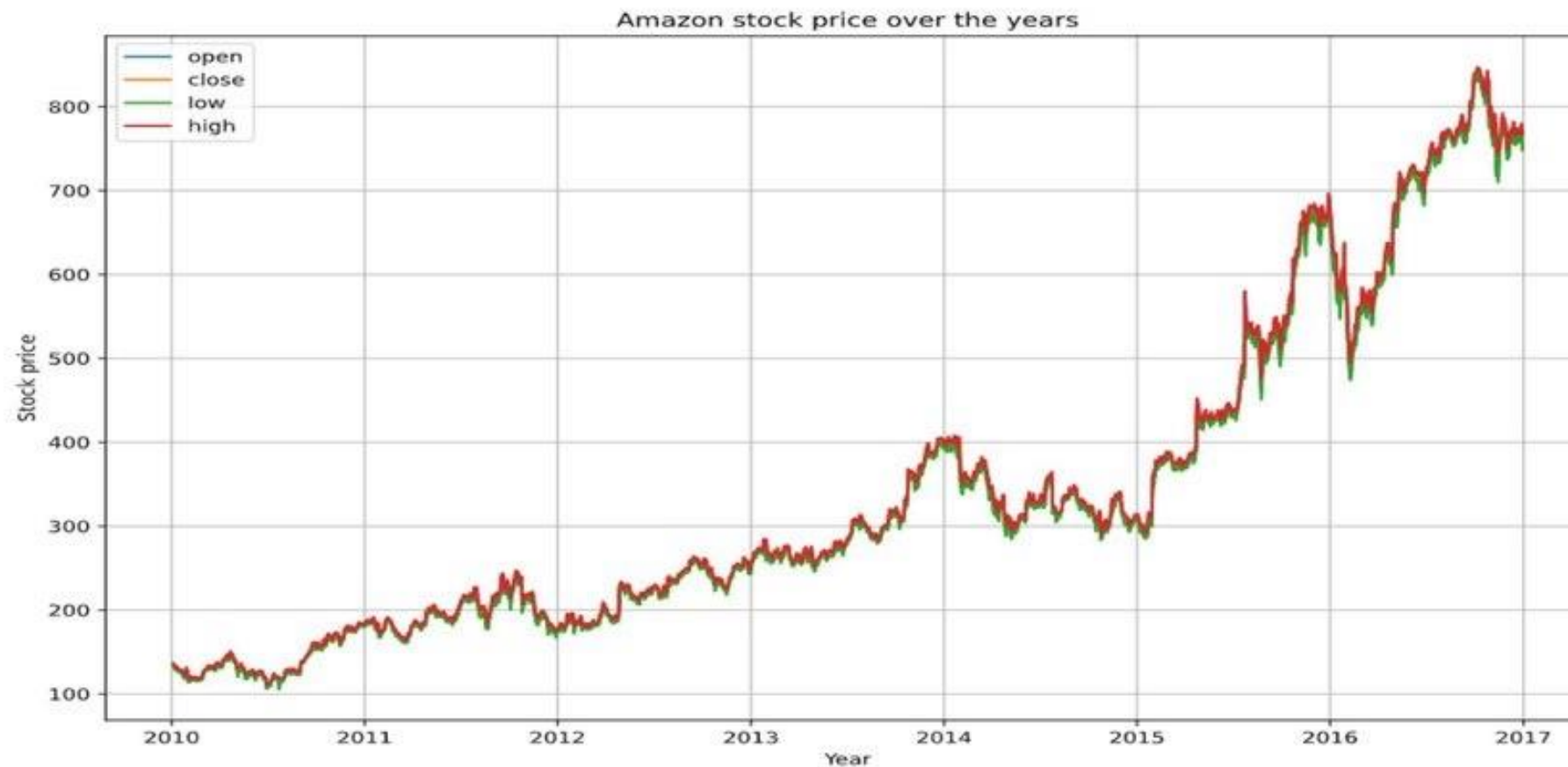
A **time series** (a sequence of data points) in data science problems



Hourly Temperature in LA on Aug 5 and 6, 2015

Examples of Vectors

Example:
(In Finance) Stock Prices



Vector Operations

Vector Addition

$$\mathbf{v} = (v_1, \dots, v_n)$$
$$\mathbf{w} = (w_1, \dots, w_n)$$
$$\mathbf{v} + \mathbf{w} = (v_1 + w_1, \dots, v_n + w_n)$$

Element-wise Operations!

Vector Multiplication

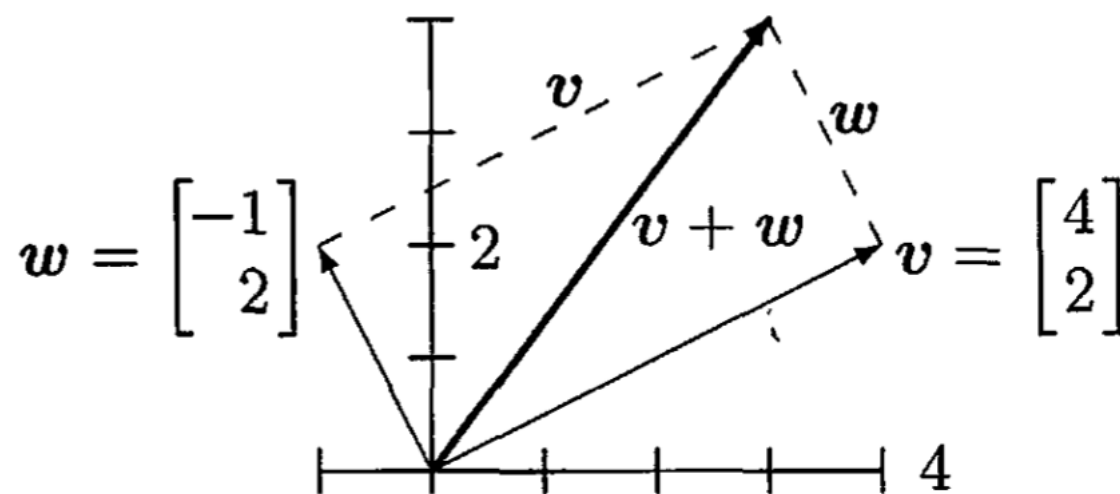
$$\mathbf{v} = (v_1, \dots, v_n)$$
$$c\mathbf{v} = (cv_1, \dots, cv_n)$$

Remark: We do not worry about the domain of the elements $v_1, \dots, v_n$ and c so far. They can be chosen from the set of real/complex numbers

Visualization of Vector Operations

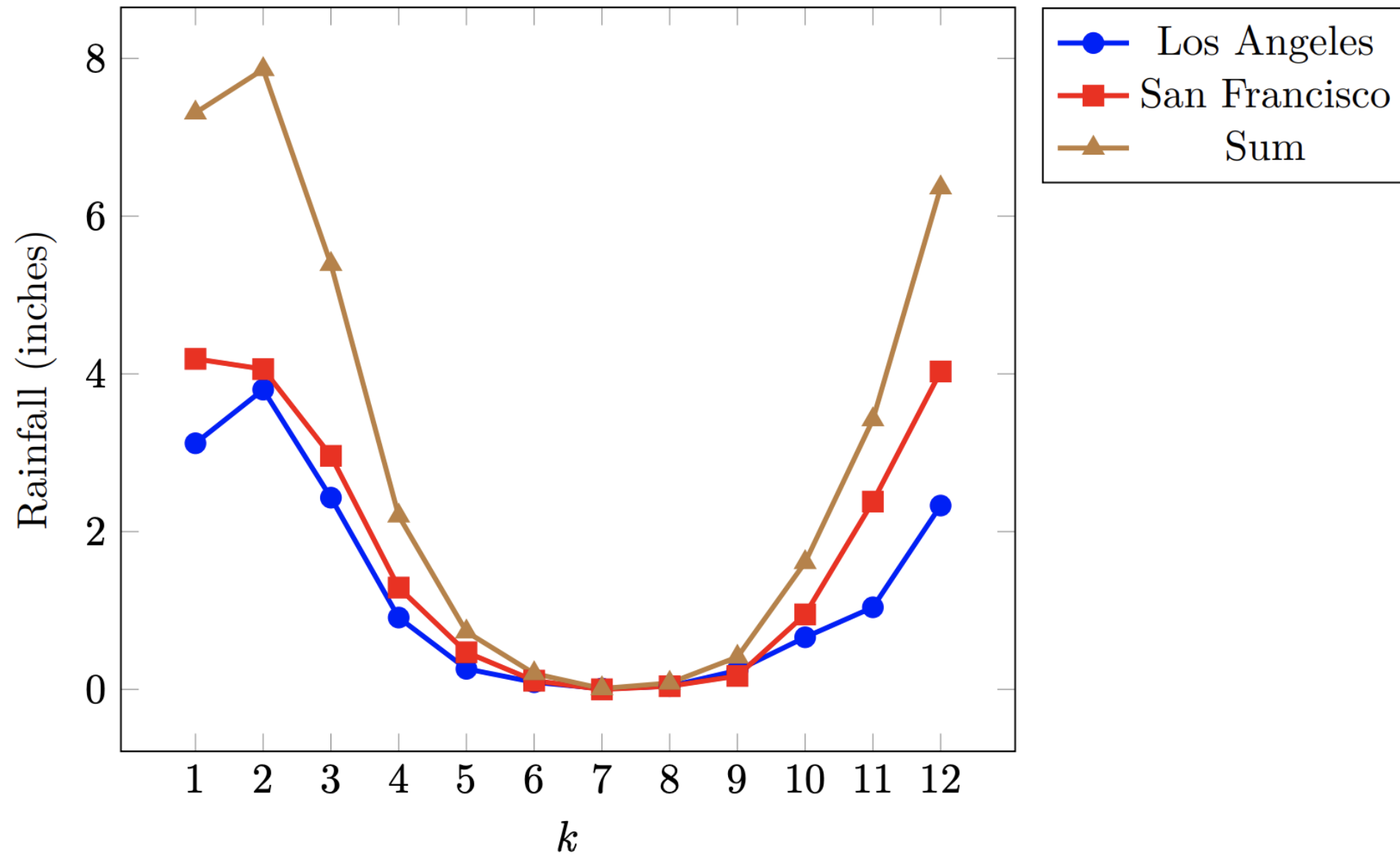
Addition $w = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ $v = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$

$$v + w = \begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$



Example of Vector Addition

Addition

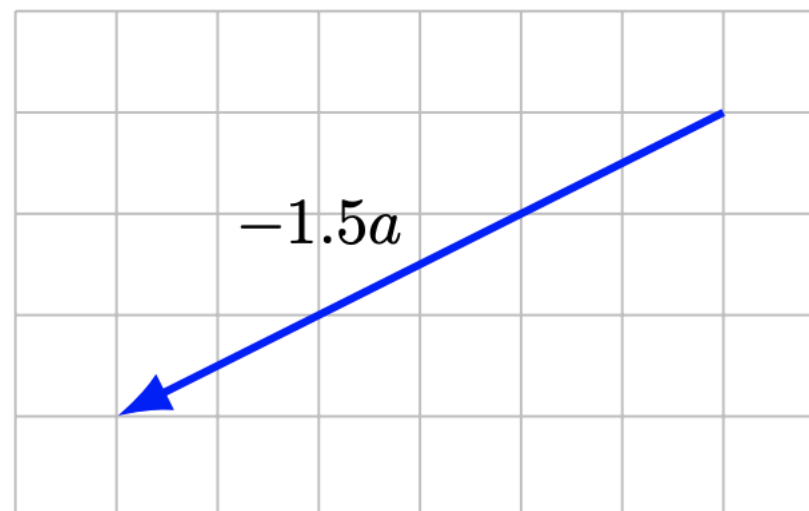
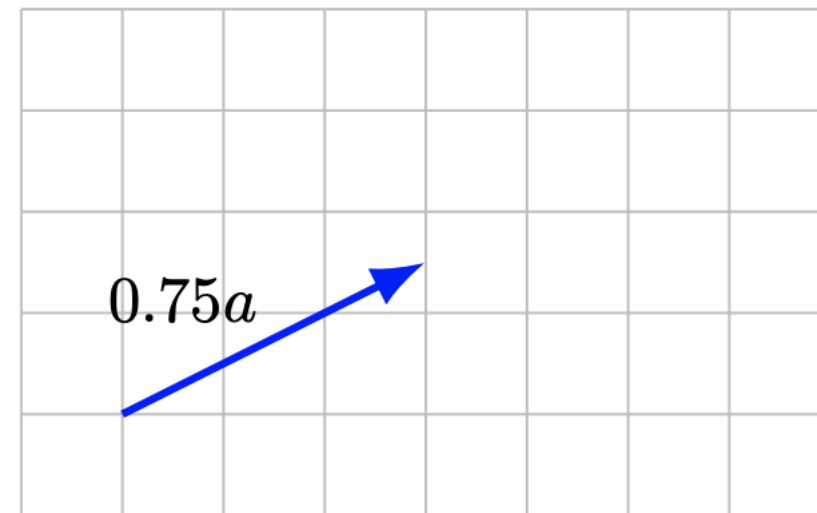
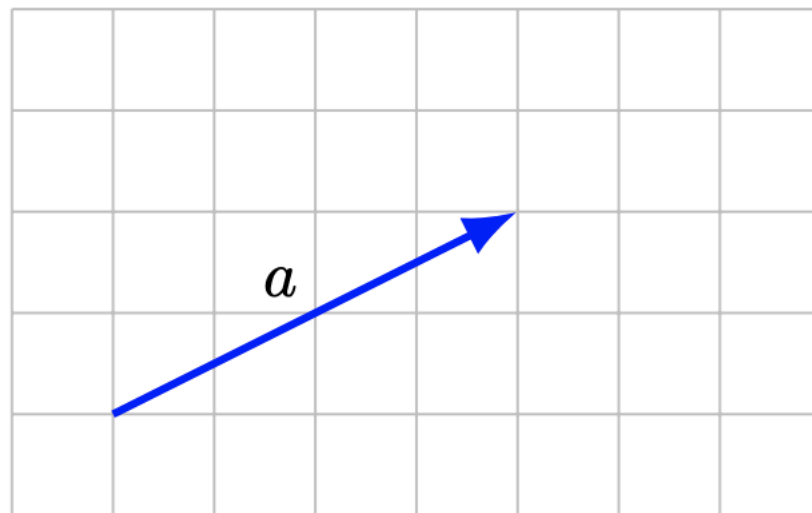


Average Monthly Rainfall in LA and SF

Visualization of Vector Operations

Multiplication (by a scalar)

Strengthen or shorten the vector; possibly changing direction (if the scalar is negative)



Vector Operations

Linear Combination

$$\mathbf{v} = (v_1, \dots, v_n)$$

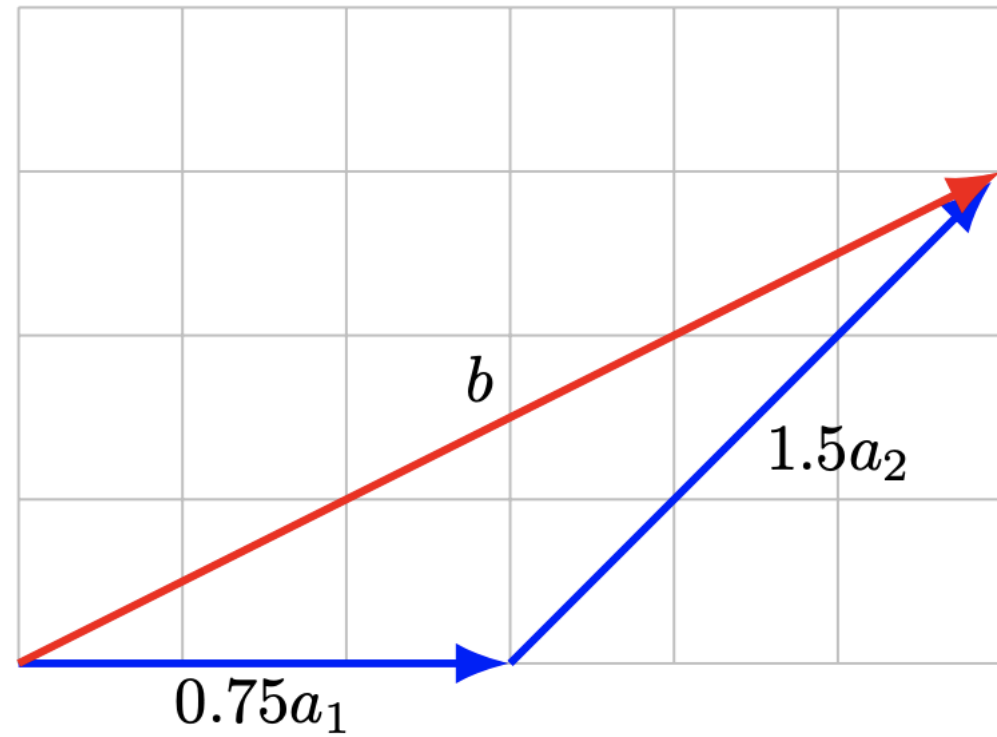
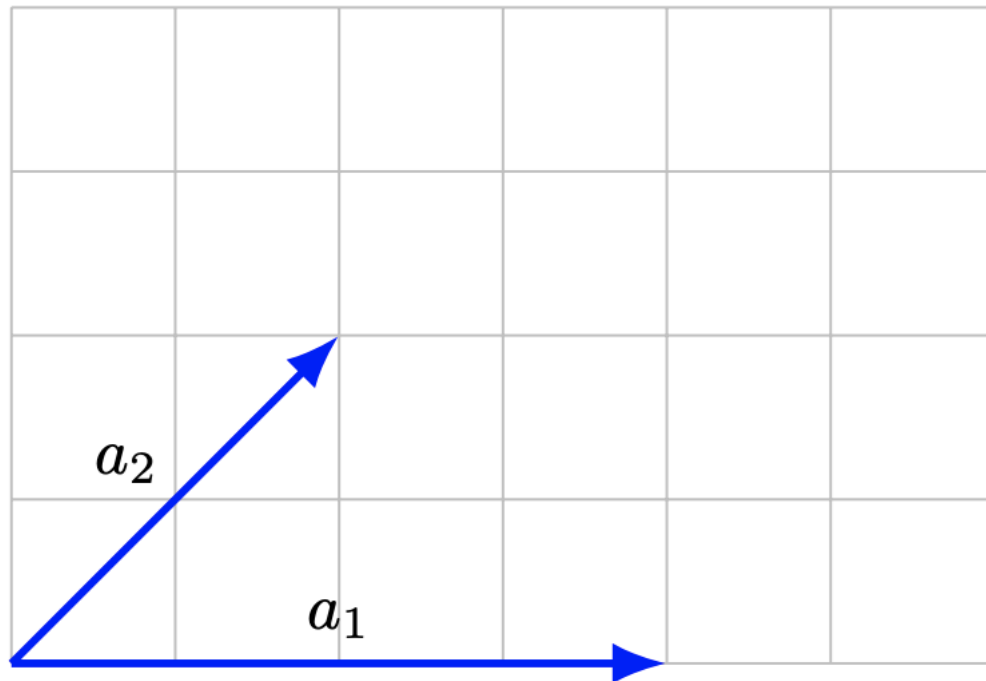
$$\mathbf{w} = (w_1, \dots, w_n)$$

$$c\mathbf{v} + d\mathbf{w} = (cv_1 + dw_1, \dots, cv_n + dw_n)$$

We only need “**addition**” and “**multiplication**” to derive the “**linear combination**” law

Visualization of Linear Combination

Linear Combination



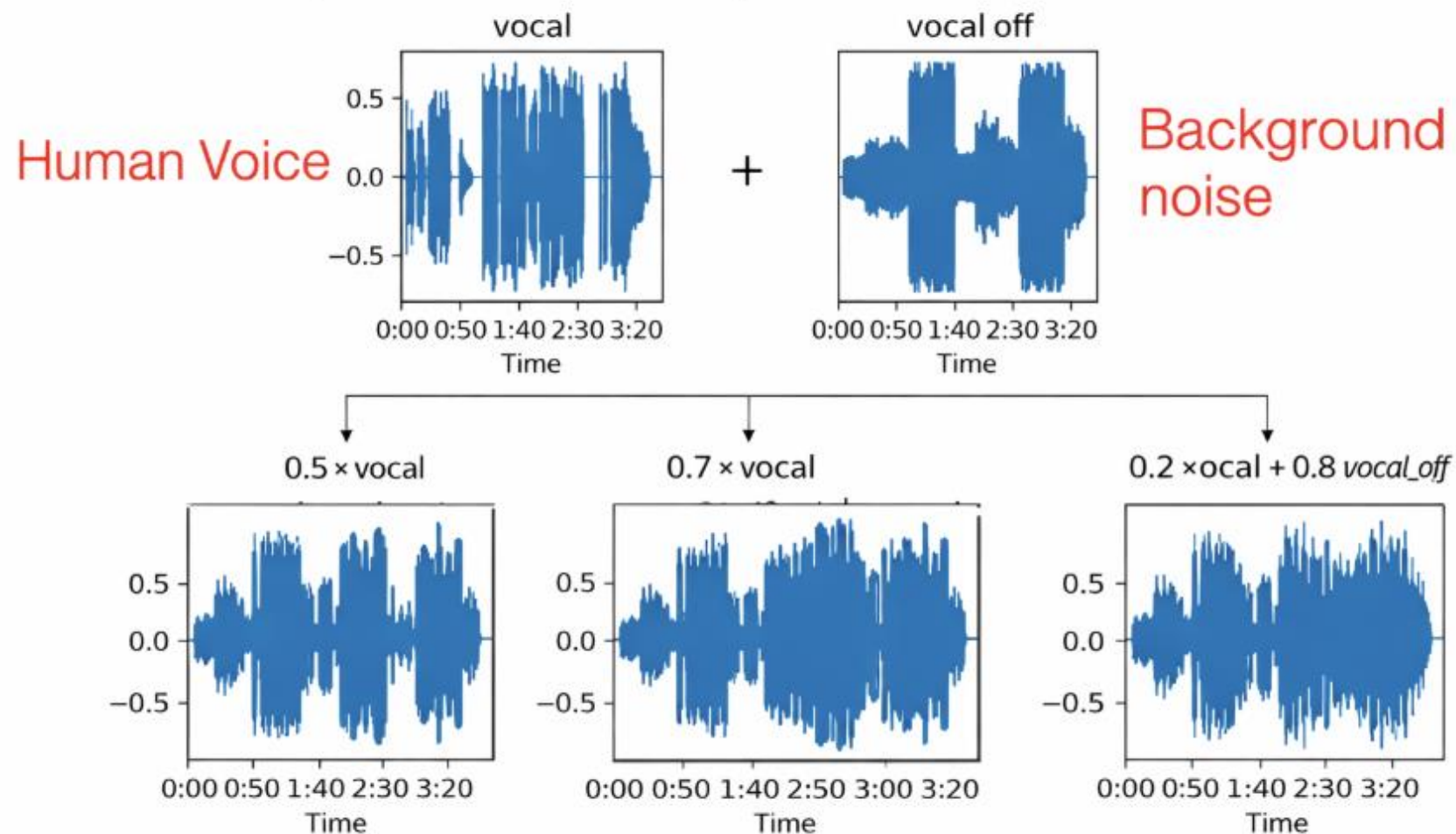
Examples of Linear Combination

Linear Combination

Example: Video/audio mixing

Linear Combination

Example: Video/audio mixing with scalar coefficients



More Examples

Example (Combinations that are not linear)

$$\mathbf{u} = (u_1, u_2)$$

$$\mathbf{v} = (v_1, v_2)$$

$$f(\mathbf{u}, \mathbf{v}) = (u_1 v_1, u_2 + v_2)$$

Combination, but NOT linear combination!

More Examples

Example (Linear Combination of Vectors)

What combination $c\mathbf{v} + d\mathbf{w}$ produces $\mathbf{u}$?

$$\mathbf{v} = (1,2)$$

Answer:

$$\mathbf{w} = (3,1)$$

$$c = 2$$

$$\mathbf{u} = (14,8)$$

$$d = 4$$

More Examples

Example (Vector Operations)

Find $\mathbf{v}$ and $\mathbf{u}$ such that

$$\mathbf{u} + \mathbf{v} = (4, 5, 6)$$

$$\mathbf{u} - \mathbf{v} = (2, 5, 8)$$

Answer:

$$\mathbf{u} = (3, 5, 7)$$

$$\mathbf{v} = (1, 0, -1)$$

Summary

In today's lecture, we have covered

- Relation of data and vector/matrices
-e.g. image as matrix; word to vectors
- Applications of linear algebra.
--engineering & science --AI and GPU.
- Tips for learning
- Definition of vectors and Vector Operations
(Slides/Notes can be found in wechat group)

(Textbook Section 1.1)

Vector Operations

Question: Can you think about any other vector operations?

The next lecture!

A General Question: What if we have three vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$

Can we write $\mathbf{x}$ as a linear combination of $\mathbf{y}, \mathbf{z}$?

The next next ...next lecture!